GEM-PIC-RZ: a Galilean electromagnetic particle-in-cell code in quasi-cylindrical geometry with free-hand grids

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We present GEM-PIC-RZ, a particle-in-cell code for laser- and beam-driven plasma accelerators in quasi-cylindrical geometry (azimuthal Fourier modes). The code implements the Galilean electromagnetic PIC (GEM-PIC) scheme, with the fast coordinate $\zeta = z - ct$ and the propagation distance $s = z$ as the slow coordinate. The electromagnetic field is split into its forward and backward characteristic components; the derivative along the slow coordinate is kept for the forward component, so the laser carrier is resolved on the ζ grid and evolves in s, while the backward and longitudinal components follow from integrals along ζ. The resulting scheme has no Courant limit: the step in s is set by diffraction and by the evolution of the plasma, not by the cell size, and can be thousands of laser wavelengths. The fields of a slice follow from two tridiagonal systems per azimuthal mode. The radial grid is staggered and non-uniform, with finite-volume operators that keep the discrete Gauss law exact, and the grids in r, ζ and s are all free-hand: the step is given at reference points and interpolated in between. We derive the numerical dispersion relation, identify a stationary odd–even mode that causes numerical Cherenkov radiation behind relativistic bunches and remove it with a filter of the beam current. The singular density where the plasma electrons close the bubble excites spurious fast modes of the reduced equations; as in the quasi-static code QUARZ, a radial smoothing of the plasma sources over a fixed length, built so that it commutes with the discrete divergence, regularizes it and makes the wake behind the bubble independent of the step. The code supports mobile ions, field ionization, particle bunches with their self-consistent initial fields at any energy, external transport elements, MPI pipelining along ζ combined with OpenMP threads, a GPU engine, and openPMD output on the native non-uniform grid. A laser wakefield benchmark against the full PIC code FBPIC agrees to 2 % in the on-axis wake, behind the bubble as well, at about 1/50 of the computing time. Over six Rayleigh lengths in a dense plasma, the depletion and red-shift of a few-cycle pulse agree with FBPIC to about 1 %, where an envelope code underestimates the depletion fivefold. Ionization injection from a nitrogen dopant gives the same injected charge, energy and spectrum as FBPIC to about 10 %. Because the carrier is resolved, the same equations also describe undulator radiation and free-electron lasers; this capability is summarized here and will be presented separately.

I. INTRODUCTION

Plasma-based accelerators driven by lasers [1, 2] or particle beams are multiscale problems: the laser wavelength $\lambda_0 \sim 1\,\mu$m, the plasma wavelength $\lambda_p \sim 10$–$100\,\mu$m and the acceleration length of millimetres to metres. Full electromagnetic PIC codes resolve λ_0 and are limited by the Courant condition, so the number of time steps is the acceleration length divided by a fraction of λ_0. Quasi-cylindrical (azimuthal-mode) codes such as Calder-Circ and FBPIC [3, 4] reduce the cost per step; the Lorentz-boosted frame [5] and Galilean coordinates with spectral solvers [6] reduce the number of steps or suppress the numerical Cherenkov instability, but the Courant limit remains.

Quasi-static codes [7–14] remove it by dropping *all* derivatives with respect to the slow time. The plasma response of a slice then follows from the sources of that slice, and the driver is advanced with a step set by its own evolution. The price is that the laser carrier can no longer be resolved: lasers enter through ponderomotive envelope models [8, 15], and radiation at the carrier frequency, harmonic generation, ionization in the instantaneous field and the emission of radiation by the beam are outside the model.

The Galilean electromagnetic particle-in-cell (GEM-PIC) algorithm [16] occupies the space between the two. It transforms the full set of Maxwell equations and the Vlasov equation to the Galilean coordinates $\zeta = z - ct$ and a slow coordinate s, either $s = ct$ (the "spatial" transformation GT1) or $s = z$ (the "temporal" transformation GT2), and splits the transverse field into its forward and backward characteristic components. The derivative along s is kept where it matters, for the forward component that carries the laser, so the carrier is resolved on the ζ grid and propagates, diffracts and is depleted and red-shifted self-consistently. It is neglected only where it appears together with the much larger ζ-derivative, for the backward and longitudinal components, which then follow from ordinary differential equations along ζ; both transformations lead to the same reduced ("universal") field system. The scheme is therefore *not* a quasi-static approximation: the slow evolution of the radiation is retained, and plasma particles that reach the next level in s stay in the window, so self-trapping is described self-consistently. Yet, like quasi-static codes, it has no Courant limit, because the field update is semi-implicit. In three dimensions, with a spectral solver in the transverse plane, GEM-PIC reproduced a 31.8 cm laser wakefield accelerator with ionization in-

jection simulated with FBPIC [16].

This paper describes GEM-PIC-RZ, the implementation of the scheme in quasi-cylindrical geometry. Its main features are: (i) the reduced field equations of GEM-PIC for azimuthal modes $m = 0, \dots, M$, discretized on a staggered, non-uniform radial grid with finite-volume operators that make the discrete Gauss law exact and decouple the slice equations into two tridiagonal systems per mode (Secs. II, III); (ii) free-hand grids in r, ζ and s, which put resolution at the axis, at the laser pulse or a witness bunch, and in a density up-ramp (Sec. III E); (iii) the numerical dispersion relation, which has no growing modes and no Courant condition, and a stationary ζ-Nyquist mode, the source of a numerical Cherenkov radiation that a beam-current filter removes (Sec. III C), and a resonance of the reduced equations that a charge-conserving radial smoothing of the plasma sources keeps from being excited at the back of the bubble (Sec. III D); (iv) a portable implementation with MPI pipelining along ζ, OpenMP and GPU engines, checkpoints and openPMD output on the native grid (Sec. IV); (v) a set of tests against analytic references and benchmarks against FBPIC, for the wake, the long-term evolution of the laser and ionization injection, and against the envelope code Wake-T (Sec. V). Since the carrier is resolved, the same equations contain the spontaneous and stimulated emission of an electron beam in an undulator; Sec. VI summarizes this free-electron-laser (FEL) capability, which will be the subject of a separate publication.

Like QUARZ [27], GEM-PIC-RZ is a *zero-touch* code: every line of its source code, of its test suite, of its benchmark scripts and of its documentation was written by an AI agent, the large language model Claude [31], working with a compiler, a shell and the test suite. The author directed the work without writing or editing a single line of code. The human contributions were those that require physical judgment: the GEM-PIC formulation itself [16], the choice of methods and of the problems to solve, physical conventions (for example, the Galilean transformation used for the particles), the criteria that decide whether a result can be trusted, and the interpretation of the results. The agent worked under a fixed rule: no feature counts as done before it passes a test against an independent analytic or numerical reference, and every change is rerun against the whole suite, so the correctness of the code rests on tests rather than on line-by-line human review. Several numerical errors described in this paper (the numerical Cherenkov radiation, the closure at the rear of the window, the self-fields of bunches with $v < c$, the undulator kick and the timing of the beam current) were found in this way, by tests that failed against their references. Every commit in the public repository [32] records the AI session that produced it, so the claim can be checked.

II. MODEL

A. Variables and field equations

We use units with $c = 1$, lengths in $1/k_0$, time in $1/\omega_0$ ($k_0 = \omega_0/c$ the laser wavenumber), densities in the critical density n_c, momenta in $m_e c$ and fields in $m_e c\,\omega_0/e$. Following Ref. [16], the fast coordinate is $\zeta = z - t$, and the slow coordinate is either $s = t$ (GT1, $\partial_t = \partial_s - \partial_\zeta$, $\partial_z = \partial_\zeta$) or $s = z$ (GT2, $\partial_t = -\partial_\zeta$, $\partial_z = \partial_s + \partial_\zeta$). The reduced field equations derived below are the same for both; the particles of GEM-PIC-RZ are advanced in GT2 (Sec. II B), so a level s of the simulation is the plane $z = s$ of the laboratory, and the slice ζ of that level is the time $t = s - \zeta$ at which the fields pass the plane. The simulation window covers $0 \le \zeta \le L_w$; its front is at $\zeta = L_w$. Every field is expanded in azimuthal modes,

$$F(r, \varphi, \zeta, s) = F_0 + 2\,\mathrm{Re} \sum_{m=1}^{M} F_m(r, \zeta, s)\, e^{im\varphi}, \qquad (1)$$

and we drop the mode index below. A linearly or circularly polarized laser lives in $m = 1$; $m = 0$ carries the wake.

The transverse fields are combined into the characteristic components

$$\begin{aligned} R_r &= E_r + B_\varphi, & L_r &= E_r - B_\varphi, \\ R_\varphi &= E_\varphi - B_r, & L_\varphi &= E_\varphi + B_r. \end{aligned} \qquad (2)$$

A wave travelling in $+z$ at the speed of light has $L = 0$ and is a function of ζ only; R is the forward (laser) field. With GT1, Maxwell's equations become, exactly,

$$\partial_s R_r = \partial_r E_z + \frac{im}{r} B_z - j_r, \qquad (3)$$

$$\partial_s R_\varphi = \frac{im}{r} E_z - \partial_r B_z - j_\varphi, \qquad (4)$$

$$(\partial_s - 2\partial_\zeta) L_r = -\Big(\partial_r E_z - \frac{im}{r} B_z + j_r\Big), \qquad (5)$$

$$(\partial_s - 2\partial_\zeta) L_\varphi = -\Big(\partial_r B_z + \frac{im}{r} E_z + j_\varphi\Big), \qquad (6)$$

$$(\partial_s - \partial_\zeta) B_z = -\frac{1}{r}\partial_r (r E_\varphi) + \frac{im}{r} E_r, \qquad (7)$$

$$(\partial_s - \partial_\zeta) E_z = \frac{1}{r}\partial_r (r B_\varphi) - \frac{im}{r} B_r - j_z. \qquad (8)$$

The forward field varies fast in ζ (on λ_0) and slowly in s (on the Rayleigh length and the depletion length). The backward field L and the longitudinal fields E_z, B_z of a co-propagating structure vary on the same ζ scales as their sources, which move with $v \approx c$, so their s-derivatives are small compared with their ζ-derivatives. With GT2 the same equations hold with $\partial_s - 2\partial_\zeta \to -\partial_s - 2\partial_\zeta$ in Eqs. (5), (6) and without ∂_s in Eqs. (7), (8) [16]. Neglecting ∂_s wherever it appears together with

∂_ζ gives, for both transformations, the reduced system

$$\partial_\zeta L_r = \tfrac{1}{2}\Big(\partial_r E_z - \frac{im}{r}B_z + j_r\Big), \tag{9}$$

$$\partial_\zeta L_\varphi = \tfrac{1}{2}\Big(\partial_r B_z + \frac{im}{r}E_z + j_\varphi\Big), \tag{10}$$

$$\partial_\zeta B_z = \frac{1}{r}\partial_r(rE_\varphi) - \frac{im}{r}E_r, \tag{11}$$

$$\partial_\zeta E_z = -\frac{1}{r}\partial_r(rB_\varphi) + \frac{im}{r}B_r + j_z, \tag{12}$$

together with Eqs. (3), (4), where $E = (R+L)/2$, $B_\varphi = (R_r - L_r)/2$ and $B_r = (L_\varphi - R_\varphi)/2$. These are the azimuthal-mode form of the universal system, Eqs. (49)–(54) of Ref. [16]: with $(x,y)\to(r,\varphi)$, $ik_x \to \partial_r$ (or $r^{-1}\partial_r r$ in the curls) and $ik_y \to im/r$, the equations agree term by term. Equations (9)–(12) are integrated from the front of the window, where $L = E_z = B_z = 0$, towards the rear. In vacuum the system gives $\omega = 2\kappa k_\perp^2/(4\kappa^2 - k_\perp^2) \simeq k_\perp^2/(2\kappa)$ for a mode $\propto e^{i\kappa\zeta - i\omega s}$, the paraxial limit of the exact $\omega = \sqrt{\kappa^2 + k_\perp^2} - \kappa$; in a cold plasma $k_\perp^2$ is replaced by $k_\perp^2 + \omega_p^2$ [16]. The backward wave is not described; Ref. [16] shows that the forward branch of the full dispersion relation approaches that of the reduced system for $\kappa \gg \omega_p$.

B. Particles

The particles follow the characteristics of the Vlasov equation in GT2 [16],

$$\frac{\mathrm{d}s}{\mathrm{d}\zeta} = \frac{v_z}{v_z - 1}, \qquad \frac{\mathrm{d}\boldsymbol{r}_\perp}{\mathrm{d}\zeta} = \frac{\boldsymbol{v}_\perp}{v_z - 1}, \qquad \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}\zeta} = \frac{\boldsymbol{F}}{v_z - 1}, \tag{13}$$

with the Lorentz force $\boldsymbol{F}$ of the fields reconstructed from the modes. A plasma particle at rest keeps its $s = z$ and crosses the whole window within one level, as in the lab frame. A particle whose s reaches the next level s^1 before it leaves the window stays inside it: it is trapped. In principle all particles can be integrated along ζ; as recommended in Ref. [16], GEM-PIC-RZ integrates the fast, trapped ones along s instead, which is explicit and more accurate. Each particle is thus in one of two lists.

Streaming particles: the plasma electrons, mobile ions and electrons freed by ionization. A fresh layer of plasma is loaded at the front of the window at every step $s^0 \to s^1 = s^0 + \Delta$, with the density $n(s^1)$ and a quiet start in r^2 and φ; one macroparticle represents the plasma of a ring of axial thickness Δ. The layer then streams through the whole window, using ζ as the independent variable, Eq. (13). Between two slices with the spacing h_n a particle is pushed over the time

$$\mathrm{d}t = \frac{h_n}{1 - v_z}, \tag{14}$$

by a two-pass Boris push [17] that makes $\mathrm{d}t$ consistent with the mean v_z over the interval; its lab position $z = s$ advances by $v_z\,\mathrm{d}t$. It deposits its current with the weight $W = q\,\mathrm{d}t/\Delta$, the charge that crosses the cell during one step [Eq. (99) of Ref. [16]]. The fields are those of the level s^1 being computed (Sec. III B).

Beam particles: bunches and trapped particles, which move with the window. They are advanced in s from s^0 to s^1 ($\mathrm{d}t = \mathrm{d}s/v_z$, $\mathrm{d}\zeta = (v_z - 1)\mathrm{d}t$) with the Vay pusher [18] in sub-steps of at most $\Delta z_{\max}$, in the fields of level s^0; the Vay pusher keeps the cancellation $\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B} \approx 0$ of a relativistic beam in its own field at any step, which the Boris pusher does not. A beam particle deposits a current with the weight q/v_z, linear in ζ.

The two lists exchange particles. A streaming particle whose lab position reaches $z = s^1$ inside the window has been trapped; it joins the beam list if $v_z \geq 0.9$, where the push along ζ becomes stiff [$\mathrm{d}\zeta/\mathrm{d}t = v_z - 1 \to 0$]. A slower one, which may happen in a strong laser field where electrons move forward by more than Δ while crossing the pulse, continues to stream in the next step from its ζ. Conversely, a beam particle that becomes slower than $v_z = 0.8$ returns to streaming. No particle is deleted, so the charge is conserved.

C. Initial fields

A laser pulse is initialized in mode $m = 1$ as a forward wave, $R_{r,1} = P^*$, $R_{\varphi,1} = iP^*$, $L = 0$, with $P = E_x + iE_y$ a paraxial Gaussian beam (with wavefront curvature and Gouy phase at a distance from focus) or a given radial profile.

A bunch of velocity β starts with its own co-moving field, $-(\nabla_\perp^2 + \gamma^{-2}\partial_\zeta^2)\psi = j_z/\beta$, $R = (1+\beta)\boldsymbol{E}_\perp$, $L = (1-\beta)\boldsymbol{E}_\perp$, $E_z = -\gamma^{-2}\partial_\zeta\psi$, with $\boldsymbol{E}_\perp = -\nabla_\perp\psi$. The problem is solved with the radial eigenvectors of the discrete Laplacian of the field solver and one tridiagonal solve along ζ per eigenvector, so a bunch in vacuum is an exact discrete equilibrium. A field that is not the co-moving one would stay behind a bunch with $v < c$ that slips back in the window and exchange energy with it; this matters for the FEL application (Sec. VI). The radial wall and the ends of the window must not cut this field; the code prints the box size it needs ($r_{\max} \gtrsim \gamma\sigma_z$, gaps of a few $\gamma^{-1}r_{\max}$ at both ends).

D. Ionization and ions

Ion species can be mobile, with their own density profile (e.g. a dopant section), and field-ionizable. The ADK tunnel rate [19, 20] of each level is evaluated with the instantaneous $|\boldsymbol{E}|$, including the laser carrier, and averaged over the two slices of the interval; ionization energies are taken from the NIST database [21]. A Monte Carlo step with exponential waiting times allows several levels per interval. The random numbers come from a counter-based hash of (seed, particle, step, slice, pass), so all back-ends draw the same numbers.

III. DISCRETIZATION

A. Radial grid and operators

The radial grid has faces ρ_i, $i = 0, \dots, N$ ($\rho_0 = 0$ is the axis), and centres c_i, $i = 0, \dots, N-1$, $\rho_i < c_i < \rho_{i+1}$. Faces carry R_r, L_r, B_z, j_r; centres carry R_φ, L_φ, E_z, j_φ, j_z. With the finite-volume measures

$$A^f_i = \rho_i(c_i - c_{i-1}), \quad A^f_0 = \tfrac{1}{2}c_0^2, \qquad A^c_i = c_i(\rho_{i+1} - \rho_i), \tag{15}$$

the operators are

$$(Ge)_i = \frac{e_i - e_{i-1}}{c_i - c_{i-1}}, \qquad (gb)_i = \frac{b_{i+1} - b_i}{\rho_{i+1} - \rho_i}, \tag{16}$$

$$(\mathrm{CE})_i = \frac{c_i E_{\varphi,i} - c_{i-1}E_{\varphi,i-1}}{A^f_i} - \frac{im}{\rho_i}E_{r,i}, \tag{17}$$

$$(\mathrm{CB})_i = \frac{\rho_{i+1}B_{\varphi,i+1} - \rho_i B_{\varphi,i}}{A^c_i} - \frac{im}{c_i}B_{r,i}, \tag{18}$$

for ∂_r to the faces and to the centres and for the two curls in Eqs. (11), (12). Two discrete identities hold exactly on any grid,

$$\mathrm{CE}\big(Ge,\ \tfrac{im}{c}e\big) = 0, \qquad \mathrm{CB}\big(\tfrac{im}{\rho}b,\ -gb\big) = 0, \tag{19}$$

because $A^f_i = \rho_i(c_i - c_{i-1})$ and $A^c_i = c_i(\rho_{i+1} - \rho_i)$; they are the discrete forms of $\nabla \times \nabla = 0$ in the two staggerings. The divergence on the measures A^c makes the discrete Gauss law exact. The particles deposit with area weighting plus a quadratic correction per interval, $w = x + \kappa x(1-x)$ with x linear in r^2, where κ is chosen so that the deposited volumes equal A^f, A^c: a uniform plasma is deposited exactly, also on a stretched grid. Fields are gathered linearly in r, which is exact near the axis for fields $\propto r$.

B. The march in ζ

The window has slices $\zeta_0 = 0 < \zeta_1 < \cdots < \zeta_{N_\zeta}$ with spacings $h_n = \zeta_{n+1} - \zeta_n$. Fields live on the slices, currents on the intervals (half-cells) between them. A step computes the level s^1 from s^0 by marching from the front slice to the rear. At slice n the slice $n+1$ of level s^1 is known.

Equations (3), (4) are integrated with the trapezoidal rule in s,

$$R^1_n = R^0_n + \frac{\Delta}{2}\left[\mathcal{F}(E_z, B_z)^0_n + \mathcal{F}(E_z, B_z)^1_n\right] - \frac{\Delta}{4}\,\mathcal{J}_n, \tag{20}$$

where $\mathcal{F}$ is the right-hand side without the current and $\mathcal{J}_n$ is the sum of the currents of the two adjacent half-cells at both levels. On a non-uniform ζ grid the current at the slice is the linear interpolation $(h_n j_{n-1/2} + h_{n-1} j_{n+1/2})/(h_{n-1} + h_n)$. Equations (9)–(12) are integrated with the trapezoidal rule over the half-cell h_n, e.g.

$$L^1_n = L^1_{n+1} - \frac{h_n}{4}\left[\mathcal{G}^1_{n+1} + \mathcal{G}^1_n\right] - \frac{h_n}{2}\,j_{n+1/2}. \tag{21}$$

R^1_n and L^1_n are linear in the unknown E_z and B_z of slice n. Inserted into the trapezoidal rules for Eqs. (11), (12), the identities (19) remove E_z from the B_z equation and B_z from the E_z equation. What remains are two independent tridiagonal systems per mode,

$$\left[16 - h_n(2\Delta + h_n)\,\nabla^2_{f,m}\right] B_{z,n} = \text{known}, \tag{22}$$

$$\left[16 - h_n(2\Delta + h_n)\,\nabla^2_{c,m}\right] E_{z,n} = \text{known} - 16 h_n\, j_{z,n+1/2}, \tag{23}$$

with the discrete Laplacians on the faces and on the centres. R and L then follow explicitly. The matrices depend only on m, Δ and h_n; they are factorized once per distinct h_n and recomputed only when Δ changes. The cost of a slice is linear in N and M.

The currents of half-cell $n+\frac{1}{2}$ at level s^1 depend on the fields of slice n, which are being computed. The march therefore uses a predictor–corrector: the predictor takes the fields of slice n from the old level, $F^1_n \approx F^0_n$ (the slow evolution in s), the streaming particles are pushed through the half-cell and deposit, the slice is solved, and the push and solve are repeated with the new fields (two passes by default). When the half-cell is final, R at slice $n+1$ is corrected for the difference between the predicted and the final current that entered its Eq. (20).

Two closures complete the scheme. At the rear slice the half-cell behind the window is set equal to the last one (zero gradient); a zero current there would halve the source of R and produce a growing spike. Near the outer radial boundary an optional absorber damps R.

C. Numerical dispersion

For a single vacuum Fourier mode $\propto e^{ik_\zeta \zeta - i\omega s}$ with transverse wavenumber $k_\perp$ ($\nabla^2_\perp \to -k^2_\perp$) on a uniform ζ grid, the scheme of Sec. III B gives

$$\tan\frac{\omega\Delta}{2} = \frac{\Delta\, h\, k^2_\perp \sin\theta}{16\sin^2(\theta/2) - h^2 k^2_\perp \cos^2(\theta/2)}, \qquad \theta = k_\zeta h. \tag{24}$$

This is the relation of the 1D plasma model of Ref. [16], Eq. (84) there, $\tan(\omega\Delta/2) = \omega_p^2 \Delta h \sin\theta / [16\sin^2(\theta/2) - \omega_p^2 h^2]$, with ω_p^2 replaced by $k^2_\perp$; the factor $\cos^2(\theta/2)$ comes from the average of the transverse fields over the two slices of a half-cell in the RZ scheme. ω is real for all k_ζ, $k_\perp$, Δ and h, so the scheme has no growing modes in vacuum, and there is no Courant condition linking Δ to h or to the radial cells. The phase $\omega\Delta/2$ passes $\pi/2$ where the denominator changes sign; with a plasma this gives the practical limit $\Delta h \le 4/\omega_p^2$ of Ref. [16], which in typical laser wakefield runs ($\omega_p^2 = 10^{-3}$, $h = 0.25$) allows Δ of order 10^4. For $\theta \to 0$ and $\Delta \to 0$, $\omega \to 2k_\zeta k^2_\perp/(4k^2_\zeta - k^2_\perp) \simeq k^2_\perp/(2k_\zeta)$, the continuum relation of the reduced system [cf. Eq. (86) of Ref. [16]] and the paraxial diffraction of the exact $\omega = \sqrt{k^2_\zeta + k^2_\perp} - k_\zeta$. The step Δ is therefore limited by accuracy: by the Rayleigh

length of the laser, the plasma and betatron evolution, and in an undulator by the period (Sec. VI).

Equation (24), like Eq. (84) of Ref. [16], also shows a defect: at the ζ-Nyquist wavenumber, $\theta = \pi$, $\omega = 0$ for every $k_\perp$. This odd–even mode is stationary in the window, like a relativistic bunch, so the macroparticle noise of the bunch current drives it resonantly; its group velocity $\mathrm{d}\zeta/\mathrm{d}s = -h^2k_\perp^2/8$ carries it slowly backwards. The result is a numerical Cherenkov radiation [22] behind the bunch, with $E_x = B_y$ (pure R) and a sign change from slice to slice. It does not act back coherently on the bunch, so it is radiation, not an instability, and it grows with the propagation distance and the shot noise; a smaller Δ does not help. A $(1, 2, 1)/4$ filter along ζ applied to the current of the beam particles, whose response $\cos^2(\theta/2)$ vanishes at $\theta = \pi$, removes the source. The plasma current is not filtered, so the wake is unchanged (Sec. V C).

D. Bubble closure and source smoothing

The continuum relation of the reduced system in a cold plasma, $\omega = 2\kappa\omega_p^2/(4\kappa^2 - \omega_p^2)$ [16], has a resonance at $\kappa = \omega_p/2$, where $\omega \to \infty$; it is the remnant of the backward branch, which the reduced system does not describe. The trapezoidal rule in s maps such modes onto modes with $\omega\Delta \to \pi$, which change sign from step to step and are not damped. In a uniform plasma of $0.002\,n_c$ they lie at $\kappa = 0.022\,k_0$, longer than the wake period, and are not excited. At the back of a bubble, however, the electrons that close it converge on the axis. For a cold plasma and a perfectly axisymmetric driver this is a caustic [9]: the density there is singular and grows without bound as the cells are refined. The local ω_p moves the resonance to resolved scales, and two numerical perturbations excite it. The first is the predicted current behind the slice in Eq. (20), whose error is proportional to Δ. The second comes from the particles that switch levels: sheath and laser-driven electrons move forward by more than Δ while they cross the window, leave their level when z reaches s^1 and continue in the next one, and each switch is a small jump of the current. Their number, and the noise, grow as Δ decreases.

In the laser wakefield case of Sec. V D both are visible behind the first bubble [Fig. 1(b)]. With $\Delta = 2.5\,\mu$m the cold-plasma run has oscillations on the scale of λ_0, with an rms deviation from FBPIC of $1.8 \times 10^{-3}\,m_e c\omega_0/e$. With $\Delta = 1.25\,\mu$m it is 9.8×10^{-3}. With $\Delta = 5\,\mu$m the field at the bubble back reaches ± 0.5, and the axis density alternates between 30 and $1800\,n_0$ from step to step.

As in QUARZ, where the same caustic limits the convergence of quasi-static runs [27], we regularize the singularity itself. Real plasmas are not cold and real drivers not perfectly cylindrical, and the spike carries no energy. After the deposit of every half-cell, the currents and densities of the streaming plasma (electrons, ions and freed electrons) are smoothed radially, per mode,

$$\begin{aligned} \boldsymbol{j}_\perp &\leftarrow (1 - a^2\nabla_{\mathrm{vec}}^2)^{-1}\boldsymbol{j}_\perp, \\ (\rho, j_z) &\leftarrow (1 - a^2\nabla_m^2)^{-1}(\rho, j_z), \end{aligned} \tag{25}$$

which gives every ring of charge a finite width a. In contrast to QUARZ, which filters every Cartesian component with the scalar operator, $\nabla_{\mathrm{vec}}^2 = \mathrm{grad}\,\mathrm{div} - \mathrm{rot}\,\mathrm{curl}$ is built from the operators of Sec. III: grad is $(G, im/c)$, the curl is CE, $\mathrm{rot}\,b = (imb/\rho, -gb)$, and $\nabla_m^2 = \mathrm{div}\,\mathrm{grad}$ is the Laplacian of the Poisson problem. Because $\mathrm{div}\,\mathrm{rot} = 0$ holds exactly on the staggered grid (with curl $= 0$ on the axis for $m \geq 1$), $\mathrm{div}\,\nabla_{\mathrm{vec}}^2 = \nabla_m^2\,\mathrm{div}$. The filter therefore commutes with the discrete divergence, and the smoothed currents conserve the smoothed charge. The vector system is pentadiagonal (j_r and j_φ interleaved) and the scalar one tridiagonal. Both are factorized once per mode, and the filter costs less than 1 % of the run time. The beam particles and the fields acting on the particles are not smoothed.

With $a = 2\,k_0^{-1}$ ($0.25\,\mu$m, $0.09\,c/\omega_p$, about the axis cell) the oscillations disappear: the rms deviation behind the bubble is 0.7×10^{-3} for both $\Delta = 2.5$ and $1.25\,\mu$m, and the run with $\Delta = 5\,\mu$m stays stable. The price is a model change of order a at the bubble back. Its position shifts slightly, the rms agreement of the first wake period changes from 1.6 to 1.9 % of the peak, and the peak of E_z at the bubble back changes from 0.047 (the spike) to 0.040 (FBPIC: 0.042). Two alternatives also suppress the oscillations: keeping electrons in their level until they have advanced several steps, and an off-centred update of R in s, which damps the fast modes by $(1-\theta)/\theta$ per step. Both are available as options. The off-centring is only first-order accurate for oscillating sources and damps the laser, so it is not suitable for long laser propagation or undulators. In a denser plasma the wake behind the bubble needs a larger a (Sec. V E).

E. Free-hand grids

All three axes use the same mapping: the step $h(x)$ is given at reference points and is linear in between, and the nodes follow from $\mathrm{d}x/\mathrm{d}k = h(x)$ with k the cell or step index. Within a segment the steps then grow geometrically and the nodes are analytic. The radial grid is fine at the axis and coarse outside; the ζ grid is fine at the laser pulse ($\gtrsim 20$ points per wavelength) or at a witness and coarse in between; the step in s is small in a density up-ramp or in an undulator and large elsewhere. All slice equations above hold for any h_n, and the beam filter acts in index space, so it removes the odd–even mode at any local spacing. A uniform grid is the special case of a constant table and is reproduced bit for bit.

IV. IMPLEMENTATION

GEM-PIC-RZ is written in C++17 without external dependencies apart from optional MPI and HDF5 libraries. The input is a structured deck with blocks for the grid, lasers, plasma, ion species, bunches, undulators and transport elements, output and numerics; all values carry physical units (or code units), and profiles of the plasma density, the laser, the bunch current and energy chirp and an undulator taper can be given as expressions, compiled once to a stack program. Unknown keys, wrong units and inconsistent settings are reported with file and line. Bunches can also be read from openPMD files of other codes, with resampling and the shot noise of the real electron number [23]. External elements (quadrupoles, solenoids, active plasma lenses, thin chicanes as R_{56} maps, planar and helical undulators) act on the particles only.

Back-ends. The serial step is the reference implementation. A parallel engine runs the same step as kernels written once as lambdas behind a thin portability layer; it compiles with a C++ compiler for OpenMP threads and with `nvcc` for NVIDIA GPUs. Fields, currents and particles stay in device memory for the whole run; per slice and pass there are three kernels (streaming push with deposition and ionization, field solve with one thread block per mode, slice preparation). The OpenMP engine reproduces the serial step bit for bit when both are compiled without FMA contraction, and an emulation of the GPU code paths on CPU threads (teams with barriers, atomic deposition, unordered appends) agrees with it to 10^{-9}. [GPU timings: the CUDA build compiles and its paths are tested in emulation; measurements on a GPU are pending.]

MPI. Within one step the march is sequential in ζ, from the front to the rear. The window is therefore split into blocks of consecutive slices, one per rank, and the ranks form a pipeline in the steps: while rank q computes step k, rank $q+1$, ahead of it in ζ, already computes step $k+1$ [24]. A rank passes to the rank behind it the boundary slice, the predicted current there, the streaming particles and the beam particles that drift across; the rank behind returns the corrected R and ghost slices for the next step. The efficiency is $N_s/(N_s+P-1)$ for N_s steps and P ranks, and the streaming plasma loads all slices equally. The ionization-injection example runs 1.90 times faster on two ranks than serially, as predicted, with identical results; the regression deck runs 1.8 times faster with two OpenMP threads. MPI combines with OpenMP threads on every rank; MPI results agree with the serial run to rounding, including runs with trapping and re-streaming of particles and restarts across rank counts.

Output. Rank 0 writes console diagnostics, bunch statistics (energy, spread, emittance, charge, also of the self-trapped charge), charge-state distributions, line-outs and maps, HDF5 files with all modes on the native grid, and checkpoints that hold the complete state including the random-number counters. A restart is exact, may change the step table in s, the output and the back-end, and may use any number of ranks. openPMD files [25] hold the fields interpolated to a uniform r–ζ grid, readable by standard tools, and in addition the exact data on the native non-uniform nodes. openPMD 1.1 has no non-uniform meshes; the native records carry their node positions as data sets referenced by a mesh attribute `gridCoordinates`, the form we have proposed, together with the QUARZ code [27], for the openPMD 2.0 standard [26].

Tests. A `ctest` suite of 25 tests runs in minutes: about 230 unit checks (parser, units, grids, deposition, atomic data, ADK rate), a regression deck for every back-end, the physics tests of Sec. V, MPI, restart, HDF5 and openPMD round trips.

V. TESTS AND BENCHMARKS

A. Verification

Table I lists the tests against analytic or independent references.

TABLE I. Verification tests (all in the test suite).

test	result
Field of an ultra-relativistic bunch in vacuum vs Gauss's law (uniform, stretched and free-hand radial grids)	E_r to 5×10^{-4}; noise $E_z<2\times10^{-3}E_r$
Same on a free-hand ζ grid (2.5–8 k_0^{-1} across the bunch)	2×10^{-3} (shot-noise level)
$\gamma=100$ bunch in its own field, 8 cm	energy kept to 2×10^{-5}
Laser focusing in vacuum ($w_0=5\,\mu$m, $z_R=98\,\mu$m, 0.4 mm)	$a_0(z)$ within 3 %
Ion collapse in a dense bunch ($n_b=100n_0$, H) [28] vs test-ion model	on-axis n_i 1–3 %; D/H delay 1.418 ($\sqrt{2}$)
Field ionization of H vs $1-\exp(-\int w\,\mathrm{d}t)$	0.02294 vs 0.02288
$N^{5+}\to N^{6+}\to N^{7+}$ ($a_0=2.2$) vs rate equations	0.9357/0.0600/0.0044 vs 0.9367/0.0593/0.0041
ADK rate vs Landau rate (H-like ions)	10^{-12}
Immobile limit of ions ($M=10^{15}$ u)	identical fields

B. Free-hand ζ and s grids

A beam-driven wake in a plasma of $0.002\,n_c$, with a driver of $\sigma_z=1.5\,k_0^{-1}$ and a witness $130\,k_0^{-1}$ behind, was run on three ζ grids (Table II). A uniform coarse grid ($h=2$) is 16 % off at the driver; a free-hand grid that is

fine ($h = 0.25$) only at the two bunches reproduces the fine uniform grid to 10^{-3} with a quarter of the slices and 3.6 times faster. In the laser wakefield example, a free-hand step in s that is small only in the density up-ramp gives the witness energy within 0.4 % of a run with uniformly small steps, with 44 instead of 80 steps; uniform large steps are 1.4 % off.

TABLE II. Beam-driven wake on uniform and free-hand ζ grids. E_z errors relative to $\max|E_z|$, witness energy change over 50 steps.

ζ grid	slices	E_z error[a]	ΔE error
uniform $h = 0.25$ (ref.)	2800	–	–
uniform $h = 2$	350	16 % / 6.6 %	1.3 %
free-hand 0.25 / 2	734	0.11 % / 0.84 %	0.06 %

[a] At the driver / at the witness.

C. Numerical Cherenkov radiation

A bunch of 1 pC at $\gamma = 1000$ ($\sigma_r = 2$, $\sigma_z = 4\,k_0^{-1}$, 16 slices) propagates 1000 steps of $\Delta = 20\,k_0^{-1}$ in vacuum. Without the filter the ζ-Nyquist part of the field energy grows to 8.6×10^{-2} of the total; with the filter it stays at 1.3×10^{-4}, the floor of the estimator. In a plasma wake ($0.002\,n_c$, 0.2 pC driver) the Nyquist fraction grows from 1.3 to 1.9×10^{-2} between 200 and 400 steps without the filter and stays at 8×10^{-5} with it, while the bunch parameters agree to four digits and the wake to 0.35 %. The filter damps structure on a scale of 31 cells per wavelength by 1 % per pass; bunches with structure of a few cells are under-resolved anyway.

D. Laser wakefield: comparison with FBPIC

We compare with the quasi-cylindrical spectral full PIC code FBPIC [4]. A laser of $\lambda_0 = 0.8\,\mu$m, $a_0 = 2$, $w_0 = 8.02\,\mu$m and a field duration of 10.6 fs (1/e half length 3.18 μm), linearly polarized and starting at focus, enters a plasma of $n_0 = 3.48 \times 10^{18}\,\mathrm{cm}^{-3}$ ($0.002\,n_c$) after a 20 μm linear ramp; the plasma column has a radius of 19.1 μm and the ions are immobile. FBPIC uses $\Delta z = \lambda_0/32$, $\Delta r = 0.2\,\mu$m, two modes and 16 particles per cell. GEM-PIC-RZ uses 100 radial cells, $0.3\,k_0^{-1}$ on the axis, stretched to $3\,k_0^{-1}$ outside, two modes, $h = 0.25\,k_0^{-1}$, $\Delta = 2.5\,\mu$m and the source smoothing $a = 2\,k_0^{-1}$ of Sec. III D. Figure 1 shows the on-axis fields. After 100 μm the first wake period agrees to 1.9 % rms of the peak field (1.6 % for a cold plasma), the minimum of E_z is -0.0335 vs $-0.0342\,m_e c\omega_0/e$, and the laser peak field 1.969 vs 1.954. Behind the bubble, the second period agrees to $0.7 \times 10^{-3}\,m_e c\omega_0/e$ rms, independently of Δ [Fig. 1(b)]. At 50 μm the two codes show different things in the up-ramp. A level s of GEM-PIC-RZ is the plane $z = s$, and its slices are the times at which the laser and the wake pass that plane (GT2), so the whole window sees the density $n(s)$ of that plane. An FBPIC snapshot at a fixed time shows the wake behind the laser at smaller z, where the density is still lower, and its wake is weaker there. The difference is one of representation, not of the physics, and it disappears where the density is constant over a window length. On one CPU core GEM-PIC-RZ took 1.7 min, FBPIC about 1.5 h.

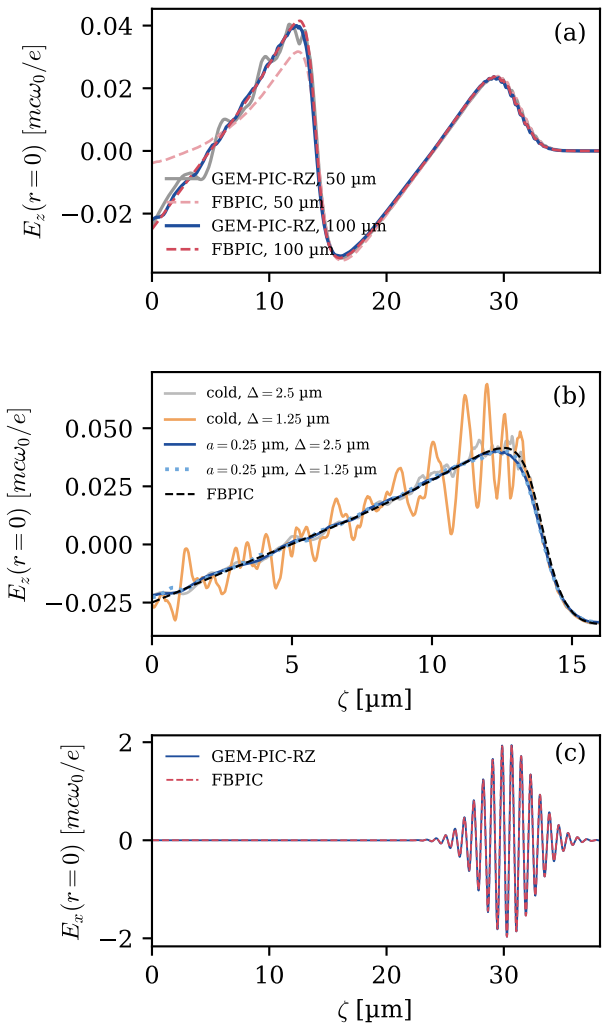


FIG. 1. Laser wakefield benchmark: (a) on-axis E_z after 50 and 100 μm of propagation, GEM-PIC-RZ with source smoothing $a = 0.25\,\mu$m (solid) and FBPIC (dashed); (b) behind the first bubble at 100 μm: cold plasma with steps $\Delta = 2.5$ and 1.25 μm, and with smoothing, Sec. III D; (c) the laser field E_x on the axis at 100 μm. The laser moves to the right.

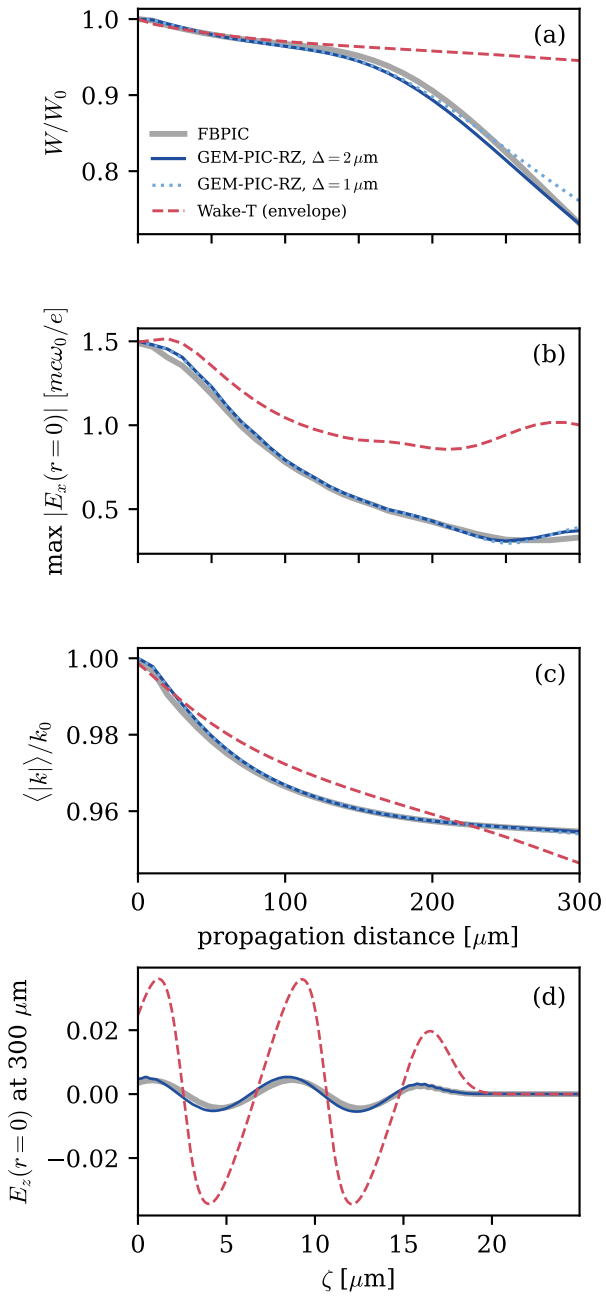


FIG. 2. Evolution of a 6.8 fs laser pulse ($a_0 = 1.5$) in a plasma of $0.01\,n_c$ over 300 μm (six Rayleigh lengths): (a) laser energy, (b) peak of $|E_x|$ on the axis, (c) spectral centroid of the laser fields; (d) on-axis E_z at 300 μm. FBPIC (grey), GEM-PIC-RZ with $\Delta = 2$ and 1 μm (blue) and the envelope code Wake-T (red, dashed).

E. Laser evolution: diffraction, depletion and red-shift

The second benchmark follows a laser pulse over many Rayleigh lengths in a dense plasma. The pulse has $a_0 = 1.5$, $w_0 = 3.56\,\mu$m ($P/P_c = 0.56$) and a field duration of 6.8 fs (2.5 laser cycles), and it starts at focus in a plasma of $0.01\,n_c = 1.74 \times 10^{19}\,\mathrm{cm}^{-3}$. Over 300 μm it diffracts, loses a quarter of its energy to the wake and is red-shifted by 4.5 %. We compare with FBPIC ($\Delta z = \lambda_0/24$, $\Delta r = 0.125\,\mu$m, two modes) and with the envelope code Wake-T [14], which combines a laser envelope solver with the quasi-static plasma model of Baxevanis and Stupakov. Figure 2 shows the laser energy (fields of the modes $m \geq 1$), the peak field on the axis, the spectral centroid of the laser fields and the wake at the end. GEM-PIC-RZ ($h = 0.2\,k_0^{-1}$, $\Delta = 2\,\mu$m) agrees with FBPIC over the whole distance: the energy to $0.013\,W_0$, the red-shift to $0.002\,k_0$, the peak field to 5 % (13 % at the end, where the pulse is weak and spread out), and the wake on the axis. With $\Delta = 1\,\mu$m the results change by less than $0.03\,W_0$. In the envelope model the pulse stays focused, drives a much stronger wake and loses only 5.5 % of its energy instead of 27 %. For this few-cycle, strongly red-shifted pulse the assumptions of the envelope model are strained; the Wake-T results do not change with resolution. GEM-PIC-RZ took 6.7 min on one core, FBPIC 3.7 h on two cores.

At this density the cold-plasma run is unstable. The fast modes of Sec. III D grow in the nonlinear wake several periods behind the laser, and E_z reaches twenty times the FBPIC value. The wake needs a smoothing length of about $0.3\,c/\omega_p$ ($3\,k_0^{-1}$). Smoothing the laser-driven $m = 1$ currents over that length changes the laser energy at 300 μm by 6 %, while for them $1\,k_0^{-1}$ suffices. The code therefore accepts a separate length for the mode $m = 0$; the run of Fig. 2 uses $a = 1\,k_0^{-1}$ ($m \geq 1$) and $3\,k_0^{-1}$ ($m = 0$). The off-centred update of R also stabilizes the wake, but it damps the laser (energy 0.58 instead of $0.73\,W_0$ at 300 μm).

F. Ionization injection

The third benchmark combines field ionization, trapping and acceleration. The laser wakefield case of Sec. V D, with $a_0 = 2.3$ and still without self-injection, contains N^{5+} ions at 1 % of the electron density. Near the peak of the pulse they are ionized to N^{6+} and N^{7+} by the ADK rate, in FBPIC as in GEM-PIC-RZ, and the K-shell electrons released there are trapped in the bubble. A level s of GEM-PIC-RZ is the plane $z = s$, and its trapped electrons are at the lab position $z = s$, while an FBPIC output at time t holds the bunch about 17 μm behind the laser, at $z = ct - 17\,\mu$m. We therefore compare at the same lab position of the bunch (Fig. 3). The ionized fraction of the dopant agrees (3.1 % N^{6+} and 0.3 % N^{7+} behind the laser). At $z = 183\,\mu$m the charge above 5 MeV is 1.58 pC against 1.76 pC in FBPIC, its mean energy 12.4 against 12.6 MeV and its rms energy spread 5.2 against 5.4 MeV, and the spectra agree up to the highest energy. Halving Δ changes the charge by 7 %. Below about 4 MeV FBPIC holds more elec-

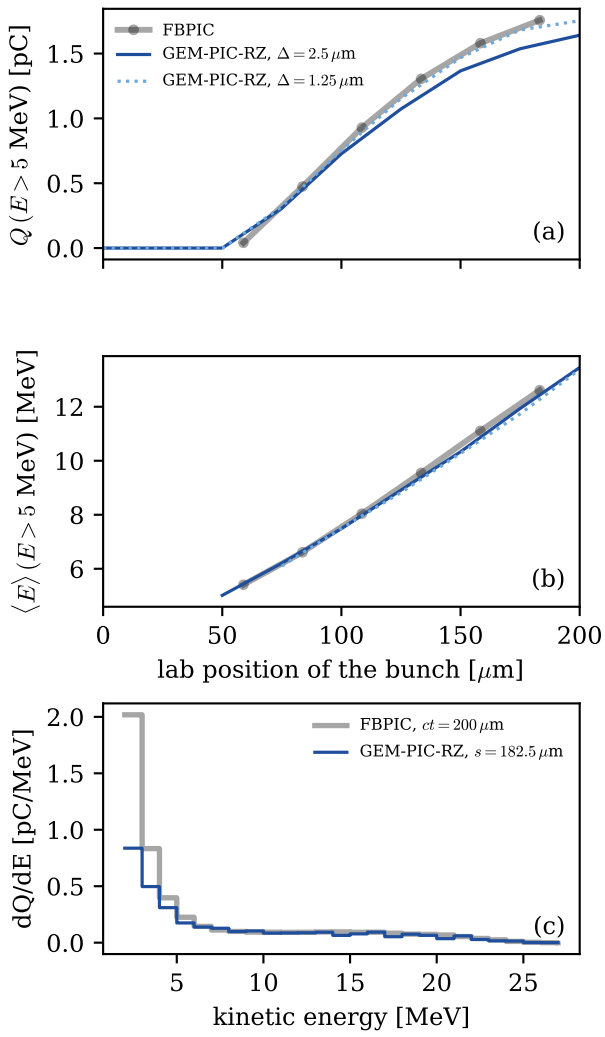


FIG. 3. Ionization injection from a nitrogen dopant ($a_0 = 2.3$, $0.002\,n_c$, 1 % N^{5+}): (a) charge and (b) mean energy of the injected electrons above 5 MeV against the lab position of the bunch, (c) their spectrum at $z = 183\,\mu$m. FBPIC (grey) and GEM-PIC-RZ with $\Delta = 2.5$ and $1.25\,\mu$m (blue).

trons: sheath electrons that are not trapped and that GEM-PIC-RZ streams through the window. Without the source smoothing of Sec. III D GEM-PIC-RZ traps 30 % more charge. GEM-PIC-RZ took 8.5 min on one core, FBPIC 3.4 h on two.

[Density down-ramp injection (a sharp ramp from 2 to 1 n_0 over $10\,\mu$m, $a_0 = 1.8$): FBPIC traps ≈ 65 pC; GEM-PIC-RZ traps 10–35 pC, depending on Δ without converging. Under investigation; not included.]

VI. UNDULATOR RADIATION AND FREE-ELECTRON LASERS

Because the carrier is resolved in ζ and the forward field evolves in s, a beam that wiggles in a static undulator field radiates through the transverse current in Eqs. (3), (4), and the radiation acts back on the beam through the gather; the radiation stays at fixed ζ while the electrons slip back by λ_r per period. Spontaneous and coherent emission and the FEL interaction are therefore contained in the same equations, with the full space-charge fields of the beam and without the period averaging and slice approximations of dedicated FEL codes [29]. The ζ grid resolves λ_r ($\lambda_r/16$ or finer), the step in s the undulator period ($\lambda_u/16$). Two details of the beam push were needed for a correct FEL coupling at such steps: the undulator field is averaged exactly over the step [a factor $\mathrm{sinc}(k_u \Delta z/2)$], and the current is deposited with the velocity at the time of the position, not half a sub-step earlier.

In brief: the coherent spontaneous emission of a short bunch in a planar undulator agrees with the far-field radiation integral to 2–4 % in energy (converging to -2 % with $\Delta\zeta$) and to 2×10^{-4} in the spectral peak; the SASE gain length of a compact FEL agrees with the Ming Xie fit [30] (6.3 vs 6.4 periods); and seeded amplifiers at 10 nm and 0.8 nm agree with GENESIS 1.3 [29] to 4 % and 0.6 % in gain length when GENESIS includes the space-charge effects that GEM-PIC-RZ contains by construction. A combined simulation of plasma acceleration, beam transport and FEL amplification in one code, and the details of these tests, will be presented in a separate publication.

VII. DISCUSSION AND CONCLUSIONS

GEM-PIC-RZ solves Maxwell's equations in the Galilean variables (ζ, s) with the slow-time derivative kept for the forward field. It resolves the laser carrier, as a full PIC code, and has no Courant limit, as a quasi-static code. Its step in s is set by the physics: hundreds to thousands of k_0^{-1} in a plasma accelerator, a fraction of the period in an undulator. The discretization on staggered, non-uniform radial nodes keeps the discrete Gauss law and $\nabla \times \nabla = 0$ exact and reduces each slice to two tridiagonal systems per mode; free-hand grids in all three directions put the resolution where it is needed. The dispersion analysis exposes a stationary ζ-Nyquist mode, and a filter of the beam current removes the numerical Cherenkov radiation it would cause, without touching the plasma response. The singular bubble closure of a cold plasma excites a resonance of the reduced equations; a radial smoothing of the plasma sources that commutes with the discrete divergence regularizes it, as in QUARZ, and makes the wake behind the bubble agree with FBPIC independently of the step. Over six Rayleigh lengths in a dense plasma, the diffraction, depletion and red-shift of a few-cycle laser pulse agree with FBPIC to about 1 % at

1/60 of its computing time, a regime in which an envelope model fails.

The model's limits follow from its ordering. Backward-propagating radiation (Raman backscatter, reflection from overdense plasma) is not described, since $\partial_s L$ is dropped. Within these limits the code agrees with a full PIC code at a small fraction of its cost, and it extends naturally to radiation by the accelerated beam. Planned work includes cross-code benchmarks for beam-driven wakes and for laser depletion, GPU performance measurements, and the FEL applications.

GEM-PIC-RZ is open source [32]; the decks and scripts of the tests in this paper are part of the repository. Its development, like that of QUARZ, shows that a production-grade simulation code can be built by an AI agent under the direction of a physicist, when every feature is held to an independent test.

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